NNN-P-001482B(DSA-DM)
October 18, 1972
Interim Revision of
Fed. Spec. NNN-P-1482A
March 19, 1970

INTERIM FEDERAL SPECIFICATION

PIPET, SEROLOGICAL, Sahli, Diluting, Hemoglobinometer, 0.02 ml.

This interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal Specification as a valid exception to Federal Specification NNN-P-001482A(DSA-DM).

1. SCOPE

1.1 This specification covers Sahli diluting pipet for use in the estimation of hemoglobin.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

FSC 6640

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS (N.B.S)

NBS 602 Testing of Glass Volumetric Apparatus.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material. The pipets shall be made from clear, high quality, milk-backing glass tubing.

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3.2 Design and dimensions. The pipet shall be of the conventional, blood-diluting Sahli design with a small uniform bore. Pipet shall be not less than 127 millimeters (mm.) or more than 140 mm. in length and shall have a milk-white background or stripe. The cross section of the pipet at any point shall be circular. The distal end shall be drawn down and the tip-end shall be ground and polished to a short taper point. The graduation mark, indicating the volume point of the capillary at 20 cubic millimeters (0.02 milliliter), shall be a fine line and shall be accurate within a tolerance of ± 2 percent of the volume, when tested as specified in 4.3.4. The graduation mark shall completely encircle the transparent portion of the pipet at right angle to the long axis and shall be located not less than 25 mm. below the top (proximal) end. Pipet shall have a slight taper at the top end so that it will fit snugly into a 1/8 inch inside diameter (ID) rubber tubing.

3.3 Identification marking. Each pipet shall be permanently marked (see 3.5) with the manufacturer's name or registered trademark, capacity (20 cubic millimeters) and an identifying serial number.

3.4 Thermal resistance.

3.4.1 Boiling. There shall be no evidence of corrosion, scumming, chipping or cracking when tested as described in 4.3.1.

3.4.2 Shock. There shall be no evidence of cracking, chipping, or breaking when tested as described in 4.3.2.

3.5 Permanency of pigment. The markings shall be permanently fused on the stem in a dark color so as to resist acids, alkalis and laboratory solvents. When tested as specified in accordance with 4.3.3, the pigmentation shall not discolor the test solution and, as judged by the eye, the appearance of all color marking shall be the same after testing as before.

3.6 Workmanship. Workmanship shall be first class throughout. Pipets shall be free of sharp edges, cracks, chips, breakage, protrusions, bubbles or other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105.

Table I. For examination

	Inspection level	AQL(percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-4	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II and MIL-STD-105.

Table II. For tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AOL
End item	Accuracy	3.2	4.3.4	S-3	*
	Boiling test	3.4.1	4.3.1	S-4	*
	Shock test	3.4.2	4.3.2	S-4	*
	Permanency of pigment	3.5	4.3.3	S-4	*

* Acceptance number shall be zero.

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4.3 Test methods.

4.3.1 Boiling test. The pipets shall be completely immersed in distilled water in a closed vessel having a steam vent and boiled continuously for 6 hours, boiling distilled water being added when necessary to compensate for evaporation losses with the pipets suspended in water so they do not come in contact with the containing vessel.

4.3.2 Shock. The pipets shall be immersed in tap water at 65° plus or minus 2.5° F., allowed to remain 10 minutes and then transferred into boiling water. Repeat the operation in 5 minutes.

4.3.3 Permanency of pigment. Immerse the pipets in a Chromic Acid Cleaning Mixture* at room temperature. Let stand for 15 minutes, then remove samples and rinse thoroughly with a cloth or cleaning tissue.

*Chromic Acid Mixture formula:

Sodium Dichromate	200 gm.
Water	1000 ml.
Sulfuric Acid	1000 ml.

4.3.4 Accuracy. Accuracy of the graduation marking (volume point) shall be tested in accordance with method outlined in the NBS Circular 602.

4.4 Examination. Pipets shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Table III. Classification of Defects

Categories	Defects
Major A	
101	Cross section at any point not circular.
102	Graduation not completely encircling transparent portion of the pipet.
103	Graduation not at right angle to the long axis.
104	Top (proximal) end not slightly tapered.
105	Distal end not ground and polished to a short taper point.
106	Marking on pipet (except manufacturer's identification) missing, not complete, correct or legible.
107	Pipet not free of sharp edges.
108	Pipet not free of chips.
109	Pipet not free of cracks or breakage.
110	Pipet not free of protrusions.

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Table III. Classification of Defects (cont'd)

Categories	Defects
Minor	
201	Pipet not free of bubbles.
202	Manufacturer's identification marking missing, not complete, correct or legible.

* Examination shall not be restricted to the defects listed herein.

4.4.2 Dimensional examination The pipets shall be examined for defects in dimension. Any dimension not within the tolerance specified herein shall be classified as a defect.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One pipet, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a spiral wound paperboard tube or in a box of appropriate size constructed in accordance with PPP-B-676, except that commercial colors are acceptable. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676, except that commercial colors are acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to the special requirements as specified in Table II of PPP-B-636. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Fiberboard boxes shall conform to the special requirements as specified in Table I of PPP-B-636. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable box is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend:

"GLASS - HANDLE WITH CARE".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
c. Federal stock number (FSN).

6.2 Medical item FSN 6640-935-1383 is the pipet covered by this specification and also replacement item for the Sahli hemoglobinometer, FSN 6630-428-1200. The pipet is suitable for use with FSN 4720-141-9070 Tubing, Rubber, Latex, 1/8 inch ID.

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0764

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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DOCUMENT IDENTIFIER AND TITLE
NNN-P-001482B(DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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